Non-stationarity and Unit Root Tests

These notes accompany Lecture 9. They combine the slide material with the explanations given in class. Terms printed in bold are defined in the glossary at the end.

Lecture 8 assumed stationarity throughout. Most economic series are not stationary, so this lecture asks four questions: what does non-stationarity mean, why does it matter, how do we detect it, and what do we do about it.

Learning outcomes – after this lecture you should be able to:

1. Distinguish a deterministic trend from a stochastic trend, and say how each is removed.
2. Define the order of integration and identify an I(0), I(1) or I(2) series.
3. Show algebraically why shocks persist when b = 1, and explain the spurious regression problem.
4. Carry out and interpret the DF, ADF, PP and KPSS tests, keeping their null hypotheses straight.
5. Apply the Box-Jenkins procedure to a non-stationary series and select an ARIMA(p, d, q) model.
6. Evaluate a forecast using RMSE and Theil's decomposition.

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1. Non-stationarity

1.1 Recap, and what fails

Covariance stationarity requires a constant mean, a constant variance, and a covariance that depends only on the lag j between two observations, not on where they sit in time. If any one of these fails, the series is non-stationary.

Three consequences follow, and each is worth stating in economic terms:

- The process has **no unconditional mean**, so nothing in the model plays the role of an equilibrium level and a long-horizon forecast has nothing to converge on. Mean reversion, and with it the statistical counterpart of long-run equilibrium, is simply absent.
- **The variance may be time-dependent,** and can grow without bound as the number of observations increases.
- For a random walk the **population autocorrelation equals one at every lag**. A finite sample will show a decline, but a gradual and nearly linear one, and that shape is the clue to look for in a correlogram.

1.2 Two kinds of trend

A non-stationary series may contain a trend, and the trend obviously changes the mean over time. But there are two quite different kinds, and confusing them leads to the wrong remedy.

Deterministic trend – trend stationary (TS)

The trend is a non-random function of time:

$$y_t = \mu + \beta t + u_t, \quad u_t \text{ i.i.d.}$$

With β = 0.45 the series climbs steadily, with random disturbances scattering observations above and below the trend line. Such processes are called trend stationary, because removing the time trend leaves a stationary series.

Stochastic trend – difference stationary (DS)

Here the trend itself is random and varies over time. The simplest case is the random walk, and adding a constant gives a random walk with drift:

$$y_t = y_{t-1} + u_t \qquad y_t = \mu + y_{t-1} + u_t$$

μ is the drift; if μ > 0 the series grows continuously.

> **Question: a random walk is a special case of what process?**
>
> An AR(1) – with the autoregressive coefficient b equal to exactly one.
>
> Write $y_t = b \cdot y_{t-1} + u_t$ and set b = 1. That single restriction is what separates a stationary AR(1) from a random walk, and it is what every unit root test is looking for.

These are called difference stationary processes, because differencing – usually once, but not always – brings them to stationary form.

The contrast, and why it matters

	Trend stationary (TS)	Difference stationary (DS)
Source of the trend	Deterministic – a fixed function of time	Stochastic – accumulated random shocks
Effect of a shock	Temporary; the series returns to the trend line	Permanent; the shock shifts the whole future path
Correct remedy	Remove the trend (detrend)	Take differences
Consequence of the wrong remedy	Differencing induces artificial MA structure	Detrending leaves the series non-stationary

1.3 Order of integration

The **order of integration** is the number of times a series must be differenced to become stationary.

Notation	Meaning
I(0)	Stationary as it stands – no differencing needed. Stationary in levels
I(1)	The first difference Δy is stationary; y itself is not
I(2)	The difference of the difference is stationary; neither y nor Δy is

Identifying processes in practice

Three real series from the lecture, as a calibration exercise:

Series	Process
Real food price index (deflated by US CPI)	Random walk
US CPI	Random walk with drift – difference stationary
Lithuanian manufacturing output	Trend stationary with a structural break

The third case is handled in EViews by interacting the trend with date ranges, using @BEFORE, @DURING and @AFTER – which lets each segment carry its own intercept and slope.

2. Consequences of non-stationarity

2.1 Shocks do not die out

Start from a general AR(1) and substitute repeatedly for the lagged term:

$$y_t = b \cdot y_{t-1} + u_t$$

$$y_t = bty_0 + u_t + b \cdot u_{t-1} + b^2 u_{t-2} + \ldots + bt{-}1u_1$$

So y_t is a weighted average of the initial value and every disturbance since, with the weight on a shock j periods old equal to b^j.

Everything turns on b:

Case	Behaviour of b^t	Interpretation
$\lvert b \rvert < 1$	$b^t \to 0$ as $t \to \infty$	The effect of a shock diminishes with time. Old shocks matter little, recent ones matter more
$b = 1$	$b^t = 1$ for all t	Every shock carries equal weight and persists forever. The series is the accumulation of all past disturbances, and the variance grows indefinitely
$\lvert b \rvert > 1$	$b^t \to \infty$	A shock becomes more influential over time – explosive, and not realistic for economic data

When b = 1 the expression collapses to a running sum:

$$y_t = y_0 + \Sigma_{i=0..t-1}\, u_{t-i}$$

The first two cases usually occur in real data. Note that stationarity requires $|b| < 1$, not $b < 1$ – an AR(1) with $b = -0.8$ is stationary, and oscillates as it decays.

2.2 Statistical consequences

- Test statistics no longer have their usual distributions – they are not normal, so conventional inference fails.
- Autoregressive coefficients are biased, so an AR fitted to a non-stationary series is estimated poorly, which damages forecasts.
- Conventional coefficient intervals no longer have their stated coverage.

2.3 Spurious regression

This is the most important consequence. Take two series with no connection whatever, each non-stationary, and regress one on the other: OLS will report a large R^2 and t-ratios that grow with the sample, both artefacts of the trending data rather than evidence of any relationship.

> **The real test**
>
> The question that settles it is whether the residuals are themselves stationary.
>
> If they are, the two series share a genuine long-run equilibrium relationship. If they are not – as in the Lithuanian GDP and Chinese CPI example – the regression is spurious, however impressive the R^2 looks.
>
> Nothing in standard OLS output warns you about this. You have to test the residuals yourself, which is the subject of the cointegration lecture.

3. Testing for non-stationarity

3.1 The Dickey-Fuller test

Begin with the AR(1) model. The random walk is the special case b = 1, and stationarity requires $|b| < 1$. So we can test whether y is a driftless random walk by testing H_o: b = 1.

It is more convenient to rewrite the equation in differences. Subtracting y_{t-1} from both sides:

$$\Delta y_t = \psi \cdot y_{t-1} + u_t, \qquad \text{where } \psi = b - 1$$

Testing b = 1 is then equivalent to testing ψ = 0. That is the Dickey-Fuller test: regress Δy on its own lagged level, and test whether the coefficient is significantly different from zero.

$$H_0: \psi = 0 \ \text{(unit root)} \qquad H_1: \psi < 0 \ \text{(stationary)}$$

Note the one-sided alternative: only ψ < 0 corresponds to stationarity, since ψ > 0 would mean b > 1.

Three specifications

Version	Equation
No intercept, no trend	$\Delta y_t = \psi y_{t-1} + u_t$
With intercept	$\Delta y_t = \mu + \psi y_{t-1} + u_t$
With intercept and time trend	$\Delta y_t = \mu + \psi y_{t-1} + \alpha t + u_t$

In all three the null is the same, but the critical values for the t-statistic on ψ differ. Dickey and Fuller produced separate tables for each – you cannot use ordinary t critical values.

> **Keep the direction of the test straight**
>
> These are unit root tests: the null is that a unit root is present.
>
> Rejecting the null is therefore evidence that the series is stationary. Failing to reject is evidence of non-stationarity.

3.2 The Augmented Dickey-Fuller test

The DF test assumes u_t is white noise. If the disturbance is autocorrelated, the test needs to allow for higher lags – which is what augmenting means:

$$\Delta y_t = \mu + \alpha t + \gamma \cdot y_{t-1} + \Sigma_{i=1..p}\, \beta_i\, \Delta y_{t-i} + u_t$$

The hypothesis is unchanged – H_0: γ = 0 against H_1: γ < 0 – but two choices must now be made:

- **Deterministic terms:** intercept, trend, both, or neither? Graph the data first, and check whether the intercept and trend are significant when included.
- **Lag length p:** the test statistic is very sensitive to this. Use AIC, SBC, or a general-to-specific approach starting from a large p and reducing it.

EViews chooses p automatically on whichever criterion you select. AIC is generally recommended here: SBC tends to pick more parsimonious models, and being too parsimonious with the lag structure risks leaving autocorrelation in the test regression – which is the very thing the augmentation exists to remove.

3.3 Why we need alternatives

Both tests lose power badly in two situations:

- Stationary processes with near-unit roots – distinguishing b = 1 from b = 0.95 is genuinely hard, especially in small samples.

- Trend stationary processes.

Low power here means a low probability of rejecting when rejection is warranted: the tests indicate that y is I(1) when in truth it is I(0). Two alternative tests address this from different directions.

3.4 The Phillips-Perron test

$$\Delta y_t = \mu^* + \delta^* t + \psi \cdot y_{t-1} + u_t$$

Here u_t is assumed I(0) but may be heteroskedastic and autocorrelated, following an ARMA(p, q). The null is again H_0: $\psi = 0$, so PP is a unit root test like DF and ADF.

The difference is in the handling of the disturbance: rather than adding lagged differences, PP corrects for serial correlation and heteroskedasticity by directly modifying the test statistic. The practical advantage is that no lag length need be specified.

3.5 The KPSS test

KPSS reverses the null. Its formulation is:

$$y_t = \beta_0 D_t + \mu_t + u_t, \quad \text{where } \mu_t = \mu_{t-1} + \varepsilon_t$$

D_t collects the deterministic components (a constant, or a constant and time trend), u_t is white noise, and μ_t is a random walk. The test is on the variance of ε:

$$H_0: \sigma_\varepsilon^2 = 0 \ \text{(y is stationary)} \qquad H_1: \sigma_\varepsilon^2 > 0$$

The logic: if $\sigma^2\varepsilon = 0$ then ε vanishes, μ_t is simply a constant, and y consists of deterministic components plus white noise – stationary. If $\sigma^2\varepsilon > 0$ the random walk component is live and feeds non-stationarity into y.

3.6 Summary of the tests

Test	Type	Null hypothesis	Rejecting the null means
DF / ADF	Unit root test	y has a unit root – I(1) or higher	y is stationary
Phillips-Perron	Unit root test	y has a unit root	y is stationary
KPSS	Stationarity test	y is stationary	y is NON-stationary

The one thing to get right

KPSS points the opposite way from the other three. A rejection by ADF and a rejection by KPSS mean opposite things.

When all four agree, they agree in this pattern: ADF and PP fail to reject, KPSS rejects – and the series is non-stationary.

Two further warnings: correct specification matters, so include an intercept and trend when appropriate; and structural breaks can make a genuinely stationary series look non-stationary if the break is not accounted for.

4. ARIMA models: estimation and diagnostics

4.1 From ARMA to ARIMA

ARMA(p, q) models can only be fitted to stationary series. If the series is non-stationary we must first make it stationary, and the traditional approach is to difference the raw data – usually after taking logs.

If the series is stationary after first differencing it is integrated of order one, I(1) – and that "I" completes the abbreviation ARIMA(p,d,q). In general:

Parameter	Meaning
p	Number of lags of the dependent variable (AR terms)
d	Number of differences required to induce stationarity
q	Number of lagged error terms (MA terms)

4.2 The Box-Jenkins procedure

Identification (steps 1–3)

1. Explore the data visually. Calculate the ACF and PACF of the raw series and judge whether it looks stationary. Justify and apply the formal tests (ADF, PP, KPSS). If stationary, jump to step 3.
2. If not stationary, take logs (where possible) and first differences, then recompute the ACF and PACF and re-apply the tests.
3. Examine the ACF and PACF of the stationary series to determine which models are good starting points.

The identification rules from Lecture 8 apply to the differenced series: an AR(p) shows an ACF dying down with a PACF cutting off after lag p; an MA(q) shows the reverse. If neither cuts off cleanly, a mixed ARMA process is suggested.

Estimation (step 4)

Estimate each tentative model and examine the coefficients. Compare using AIC and the Schwarz Bayesian criterion, choosing the smallest values. Of the two, SBC is preferable because it favours parsimony.

Diagnostic checking (steps 5–6)

- Check that the coefficient on the longest lag is significant. If it is not, you probably have too many parameters and should reduce p or q.
- Check the ACF and PACF of the residuals. If the model has enough parameters, all residual autocorrelations will be insignificant. The Ljung-Box Q-statistic gives the joint test.
- Compare AIC, SBC and adjusted R^2 across models to identify the parsimonious one.
- Plot the residuals, look for outliers and for periods where the model fits poorly. Avoid overfitting.

If changes are needed, return to the estimation step and iterate.

4.3 Worked example: Lithuanian core CPI

The series is the Lithuanian CPI excluding energy and unprocessed food, monthly.

Step 1 – is it stationary?

The plot shows a clear trend, and the ACF does not die down – both point to non-stationarity. The formal tests confirm it, and note how the three verdicts line up:

Test	Result	Conclusion
ADF	Fail to reject H_0 (unit root)	Non-stationary
Phillips-Perron	Fail to reject H_0 (unit root)	Non-stationary
KPSS	LM-stat 0.380 > 0.216 (1% critical value) – reject H_0 (stationarity)	Non-stationary

For KPSS, remember that rejecting means rejecting stationarity. All three tests therefore agree: the series is non-stationary.

Step 2 – transform

Take logs, then first differences. The differenced series looks stationary and the tests confirm it. Multiplying the log difference by 100 gives an approximate monthly growth rate in percent.

Step 3 – identify candidates

The CPI is I(1), so we model the first difference. The ACF decays gradually (AR-type behaviour) while the PACF dies down after lag 3. The natural first choice is ARIMA(3,1,0); ARIMA(1,1,1) is also plausible and worth checking, particularly if the residuals of the first model show autocorrelation.

Steps 4–6 – estimate and check

Model	Findings
ARIMA(3,1,0)	Coefficients significant, R^2 adequate, no visible residual autocorrelation. One outlier at the Global Financial Crisis
ARIMA(1,1,1)	Coefficients significant, no residual autocorrelation, and SBC slightly lower than ARIMA(3,1,0) – the preferred model
ARIMA(1,1,1) with crisis dummy	No significant changes, so the outlier is not driving the results

4.4 Forecast evaluation

Use a dynamic forecast for an iterative (multi-step) path, and compare with the actual data. Two families of measure matter.

$$RMSE = \sqrt{[\ (1/n)\ \Sigma\ (\hat{y}_t - y_t)^2\ ]}$$

RMSE (root mean squared error) is the typical size of the forecast error, in the units of the variable itself – here about 0.135, in the same units as the monthly inflation rate. Because errors are squared before averaging, RMSE penalises large misses more heavily than the mean absolute error (0.106), so RMSE ≥ MAE always. A wide gap between them signals a few relatively big errors.

Lower is better, but RMSE is most useful for comparing rival models on the same variable and sample, not as a pass/fail number on its own.

Measure	Value	Reading
Theil inequality coefficient	0.21	Reasonably accurate. U = 0 is a perfect forecast, U = 1 the worst possible
Bias proportion	0.03	Negligible – no systematic over- or under-prediction
Variance proportion	0.83	Dominates the error: the forecast is too smooth and fails to track inflation's swings
Theil U2	0.63	Below one, so the model still beats a naïve no-change benchmark

What the decomposition is telling you

The forecast is essentially a flat line near the mean while the actual series swings around it.

That is not a failure of calibration – the bias is almost zero. It is a failure to capture variability, which is exactly what a high variance proportion means.

This is also the honest limitation of univariate modelling: with only the past of the series to work with, there is nothing to predict the swings from. Multivariate models are the next step.

Key terms

Term	Meaning
Non-stationary	Failing any of the three covariance-stationarity conditions: constant mean, constant variance, lag-dependent covariance.
Deterministic trend	A non-random function of time. Removed by detrending.
Trend stationary (TS)	Stationary once a deterministic trend is removed. Shocks are temporary.
Stochastic trend	A random trend that varies over time; the accumulation of shocks.
Difference stationary (DS)	Stationary after differencing. Shocks are permanent.
Random walk	$y_t = y_{t-1} + u_t$. An AR(1) with b = 1.
Drift	The constant μ in $y_t = \mu + y_{t-1} + u_t$, producing continuous growth.
Order of integration	The number of differences needed to induce stationarity; I(d).
Unit root	The case b = 1, where shocks persist indefinitely and the variance grows without bound.
Spurious regression	An apparently significant relationship between unrelated non-stationary series. Diagnosed by testing whether the residuals are stationary.
Dickey-Fuller test	Regress Δy on its lagged level and test $\psi = 0$. Null: unit root.
Augmented Dickey-Fuller	DF plus lagged differences, to handle autocorrelated disturbances.
Phillips-Perron test	Unit root test correcting for serial correlation and heteroskedasticity by adjusting the statistic; no lag length required.
KPSS test	A stationarity test – the null is that y is I(0). Rejection means non-stationarity.
Low power	A low probability of rejecting a false null; the reason DF/ADF struggle with near-unit roots.

Term	Meaning
ARIMA(p, d, q)	AR order p, d differences, MA order q.
RMSE	Root mean squared forecast error, in the units of the variable.
Theil decomposition	Splits forecast error into bias, variance and covariance proportions.

Review questions

Use these to check your understanding before the seminar.

1. State the three conditions for covariance stationarity, and give an economic interpretation of what is lost when the first one fails.
2. Distinguish a deterministic from a stochastic trend. For each, state the correct way to induce stationarity and what goes wrong if you use the other method.
3. Show that a random walk is an AR(1) with b = 1. Why is this single restriction the target of every unit root test?
4. Write out y_t in terms of y_0 and all past disturbances. Explain how the weight on a shock j periods old depends on b.
5. For b = 0.9, b = 1 and b = 1.05, describe what happens to the influence of a shock as time passes. Which cases occur in real economic data?
6. Why is $|b| < 1$ the correct condition for stationarity rather than $b < 1$? Give an example of a stationary AR(1) with a negative coefficient.
7. A regression of Lithuanian GDP on Chinese CPI gives $R^2 = 0.9$ and a t-statistic of 12. What further test would you run before believing it, and why does standard OLS output not warn you?
8. Derive the Dickey-Fuller regression from the AR(1), and explain why testing b = 1 is equivalent to testing $\psi = 0$.
9. Why is the DF alternative one-sided? What would $\psi > 0$ imply about b?
10. What problem does the ADF test solve relative to DF? Write the ADF regression and identify which coefficient is tested.
11. Why is AIC generally recommended over SBC when choosing the ADF lag length? Relate your answer to what the augmentation is for.
12. How does Phillips-Perron handle autocorrelated errors differently from ADF? What practical advantage follows?
13. State the KPSS null and explain the logic: why does $\sigma^2\varepsilon = 0$ imply stationarity?
14. ADF gives p = 0.60 and KPSS rejects at 1%. Are these results consistent? What do you conclude about the series?
15. The Lithuanian CPI is I(1). Set out the full Box-Jenkins sequence you would follow from raw data to a chosen model.
16. ARIMA(3,1,0) and ARIMA(1,1,1) both have significant coefficients and clean residuals. On what basis was ARIMA(1,1,1) preferred?

17. The forecast has bias proportion 0.03 and variance proportion 0.83. Describe in words what the forecast path looks like, and explain why this is a limitation of univariate modelling rather than a mistake.

References and further reading

Asteriou, D., Hall, S.G. Applied Econometrics – Chapters 13 and 16.

Macroeconometric Forecasting. IMF online course. URL: https://www.edx.org/course/macroeconometric-forecasting-2.